Reddit for Financial Insight: A Comparative Study with Real-Time Stock Data

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Abstract

Social media platforms have increasingly become influential in shaping investor decisions, notably within financial communities such as Reddit's stock discussion forums. This study investigates the relationship between stock market discussions on Reddit, specifically the sentiment and topical content, and actual stock price movements, focusing primarily on Tesla (TSLA). Utilizing data collected from Reddit spanning from January 2022 to March 2025, coupled with corresponding stock price data from Yahoo Finance, we employ sentiment analysis, topic modeling (Top2Vec), and statistical testing methods such as Granger causality analysis. Our findings indicate that Reddit sentiment does not consistently predict future stock price movements, suggesting that discussions are largely reactive rather than predictive. However, event-based topic analysis reveals specific instances where significant discrepancies between investor expectations and actual stock outcomes occur. These discrepancies provide valuable insights into market behavior, highlighting moments of collective misunderstanding or hype among investors. This research contributes to the ongoing discourse on the reliability and impact of social media-driven market sentiment, demonstrating the complexity and contextual sensitivity required in interpreting online financial discussions.

1 Introduction

Social media has dramatically changed how investors research, strategize, and respond to market activities. Sites such as Reddit and their stock-focused subreddits like r/stocks or r/investing have accumulated a huge following as they provide fast market sentiment that reactions in the market can be forwarded instantly. Nowadays, many of these discussions on social platforms are used to make trading decisions, as many of the users have begun to view such social interactions as precursors or clues for the upcoming of the market direction.

Regardless of this matter, capturing the sentiment of social media and using it for analysis remains controversial. Existing literature captures this debate without consensus; some claim that online interaction, including performing sentiment analysis on social media, offers useful guidance for predicting stock prices, while others argue that social media commentary is too detached from the actual events and do not lead the market. This uncertainty highlights a critical gap: Do social media outlets like Reddit provide useful foresight toward market behavior, or are they simply reactive channels that can mislead investors with unsubstantiated claims and capture the markets after they've changed?

This study aims to fill this research gap by conducting an in-depth analysis of Reddit discussions specifically related to Tesla (TSLA) stock over a period from 2022 to 2025. By leveraging advanced text mining techniques, sentiment analysis, and topic modeling, we seek to understand the relationship between Reddit sentiment and Tesla's actual stock price movements. Furthermore, we apply rigorous statistical methods—including Granger causality and decision-tree modeling—to test whether Reddit discussions truly precede significant market movements or merely echo them afterward.

2 Related Work

Bollen, Mao & Zeng (2011), This Study examined several million public tweets to derive daily mood indices such as "Calm", "Happy", and "Anxious" —and demonstrated that certain aggregated mood dimensions could predict movements in the Dow Jones Industrial Average up to a week in advance. Their study is foundational for social-media sentiment research because it shows that carefully aggregated online mood can carry genuine market information. We build on their methodology of large-scale sentiment aggregation but find that, on Reddit, sentiment constantly lags Tesla's price moves rather than leading them—highlighting important platform-specific differences in predictive power.

Yuan (2025) conducted a case study on the GameStop short squeeze to analyze how social-media sentiment shapes investor behavior, blending behavioural economics frameworks with sentiment analysis. The paper finds that collective hype can drive extreme volatility, but that much of the chatter arrives after major price inflections. This mirrors our Tesla findings: Reddit sentiment spikes around key events (e.g., earnings, rumors) but only after prices have already swung, reinforcing the view that unfiltered Reddit discussions are reactive and occasionally contrarian.

Tetlock (2007), Paul C. Tetlock's study "Giving Content to Investor Sentiment: The Role of Media in the Stock Market" (Journal of Finance, 2007) analyzes how the tone of newspaper columns (measured via a simple lexicon-based sentiment score) predicts stock-market movements. He finds that high levels of pessimism in the media precede downward price pressures, while optimism tends to follow market rallies. This work is relevant because it shows that even traditional news sentiment can have predictive value when properly quantified. In contrast, our Reddit-based sentiment—though similarly quantified via advanced models—fails to anticipate Tesla's returns and instead lags price moves, highlighting how the choice of data source and sentiment methodology can critically affect forecasting outcomes.

Chen, De, Hu & Hwang (2014) analyze millions of "trade opinions" on StockTwits, a social platform where users explicitly post bullish or bearish signals and show that aggregated crowd sentiment can significantly predict next-day stock returns even after accounting for traditional market indicators. This work is directly relevant because it demonstrates that, in a structured environment with clear buy/sell cues, social-media sentiment can lead price movements. By contrast, our findings on Reddit, discussion-focused platform reveal that unfiltered crowd sentiment constantly lags Tesla's price moves and adds no forecasting power, underscoring how platform design and the clarity of user signals fundamentally shape the predictive value of social-media data.

Sul, Dennis & Yuan (2017) analyze 2.5 million tweets about S&P 500 firms and show that sentiment from low-follower users—particularly those whose posts aren't retweeted—significantly predicts next-day, 10-day, and 20-day stock returns, even supporting an 11–15 This work demonstrates that, on Twitter, platform mechanics and user-level filtering can surface truly forward-looking signals. In contrast, our analysis of Reddit's more open, unfiltered discussions around Tesla finds that crowd sentiment constantly lags actual price moves and provides no reliable forecasting edge, underscoring how the choice of social-media venue and signal-extraction method crucially shapes sentiment's predictive power.

3 Data Collection

To address our research question, we compiled an extensive dataset focusing on Reddit discussions related to Tesla's stock (TSLA). Reddit, known for its active communities and candid discussions, provides a valuable window into public sentiment, market predictions, and investor psychology. We specifically targeted popular finance-related subreddits, including r/stocks, r/StockMarket, r/WallStreetBets, r/Investing, and r/finance, due to their large user base and frequent discussions on stock-related events.

We employed a systematic approach to data collection using Reddit's API through the PRAW library (Python Reddit API Wrapper). Our collection strategy involved extracting Reddit posts and comments containing the keywords "Tesla," "TSLA," "Elon Musk," "stock," and "shares." Data was gathered over a span of three years and three months—from January 1, 2022, to March 31, 2025—to ensure we captured a broad range of market events, trends, and user sentiments over time.

Furthermore, to evaluate the relationship between public sentiment and stock price movements, we gathered daily Tesla stock price data for the same period from Yahoo Finance using the yfinance Python library (Aroussi, 2019).

In total, our data collection yielded 208,785 posts and comments. Data was systematically collected and segmented by year, and then combined into a unified dataset for robust analysis. To ensure data quality and relevance, a preprocessing phase was conducted to remove generic or non-informative comments. Specifically, we filtered out comments consisting solely of emojis or generic expressions such as 'yes', 'agree', 'thanks', and similar sentiment-neutral phrases, since these do not contribute meaningful insights to sentiment or topic modeling analyses. Through this preprocessing step, a total of 1,277 entries were removed—comprising 463 generic comments, a single missing value and 813 emoji-only or symbol-only comments. The final cleaned dataset thus consists of 207,508 informative entries providing a reliable foundation for subsequent in-depth sentiment analysis, topic modeling, and correlation with stock price movements.

To facilitate sentiment and topic analysis, text preprocessing was performed, including tokenization, stop-word removal, and the cleaning of special characters. Each Reddit post and comment was further analyzed using the RoBERTa model for sentiment classification, generating sentiment scores ranging from negative to positive.

4 Methods

In order to assess the reliability and value of Reddit discussions in relation to stock market trends, particularly those surrounding Tesla (TSLA), we developed a multi-phase analysis pipeline combining natural language processing, topic modeling, and statistical modeling. Our approach was driven by the goal of identifying whether Reddit user discussions could meaningfully signal or influence stock price changes, or if they primarily reacted to events post hoc.

4.1 Sentiment Analysis with RoBERTa:

For sentiment classification, we used the cardiffnlp/twitter-roberta-base-sentiment model, a RoBERTa-based transformer model (Devlin et al., 2019) finetuned on Twitter data. This choice was intentional and methodologically strategic for several reasons. Firstly, the linguistic style of Reddit shares notable similarities with Twitter: both platforms are informal, often laden with slang, sarcasm, and emotionally charged short posts. A model trained specifically on Twitter content is therefore inherently better suited to capture the nuanced sentiment of Reddit discourse compared to models trained on formal corpora like movie reviews or news articles. Second, RoBERTa (A Robustly Optimized BERT Pretraining Approach) is known to outperform vanilla BERT in many text classification tasks due to its dynamic masking and removal of Next Sentence Prediction objective.

The CardiffNLP model further fine-tunes this robustness by optimizing sentiment classification for social media content. It outputs sentiment polarity on a scale from -1 (strongly negative) to 1 (strongly positive), along with a confidence score. This granularity makes it particularly well-suited for our use case: extracting and quantifying sentiment in financial discussions around volatile stocks like Tesla.

4.2 Event-Based Topic Modeling with Top2Vec

Understanding public discussions on social media and their relation to stock-price fluctuations has become a growing area of interest in financial computing and social-data analysis. One of the recent advancements in topic modeling is the Top2Vec algorithm (Angelov, 2020), which learns joint embeddings of documents and words to detect topics automatically. Unlike traditional LDA, Top2Vec requires no prior choice of topic count, relying instead on semantic similarity in the embedding space.

Top2Vec is especially useful for unstructured text such as Reddit comments, whose style, tone, and vocabulary shift rapidly. Embedding-based clustering captures subtle relationships that would evade bag-of-words models and pre-defined taxonomies.

However, applying Top2Vec to a multi-year corpus diluted the specific contexts around market shocks, surfacing thousands of diffuse topics. We therefore adopted an *event-based* approach: Top2Vec was run only on Reddit posts from high-impact dates where Tesla's price moved sharply—either an unexpected dip or a sudden rally.

Table 1: High-impact event windows analysed with Top2Vec and time-lag tests. Each narrative lists *all* keywords returned for the dominant topic on the given date.

Date	Dominant Reddit narrative (Top2Vec keywords)	Posts
2022-03-16	*lockheed, congress, plummeting, fewer, martin, march, wednesday, close, year, highs*	180
2022-04-21	*florida, disney, bill, governing, house, passes, netflix, investors, stock, cramer, percent, down, buy*	133
2023-01-03	*jpmorgan, epstein, island, lawsuit, bank, hedge, scandal, trader, risk, fund*	85
2023-04-20	*seagate, huawei, blacklisted, penalty, billion, fine, export, violation, chips, embargo*	82
2023-09-13	*supervillain, superhero, musk, elon, ceo, working, hardest, meme, twitter, electric*	158
2024-06-07	*gamestop, trader, retailer, roaring, kitty, tumbles, shares, short, squeeze, meme, rally*	364
2024-12-11	*nintendo, happened, wtf, leak, rumor, switch, mario, speculation, surprise, thread*	290
2025-02-14	*paywall, rddt, content, lock, ceo, company, subscription, spy, hathaway, voo, tesla*	631
2025-03-05	*subsidy, tariffs, chips, trump, act, billion, automakers, delays, tesla, stock, sales, export*	696

Figure1 illustrates a representative window: a five-day span around 13 Sep 2023 where sentiment spiked the day *after* the price peak, suggesting reactive rather than predictive chatter. Similar timing patterns appeared in other windows.

By isolating Reddit discussions on these dates and applying Top2Vec only to the relevant comments, we obtained compact, interpretable topic clusters that reflected the core narratives of each event. Using the *Universal Sentence Encoder* for embeddings and hierarchical topic reduction, we distilled dominant themes into a manageable set per date. This event-based focus clarified whether Reddit narratives offered forward-looking insight or merely post-hoc commentary, enabling a deeper comparison between sentiment scores and real-world market outcomes.

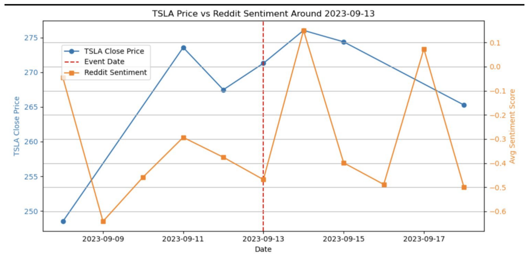

Figure 1: Tesla Stock Price vs Sentiment Score around 20-23-9-13

4.3 Statistical Tests for Temporal Sentiment Dynamics

To determine whether Reddit sentiment exhibits systematic calendar-based patterns, we first applied a one-way ANOVA across the seven weekdays. The ANOVA assesses whether the between-weekday variance in mean sentiment is significantly larger than the within-weekday variance, as measured by the F-statistic:

$$F = \frac{\mathrm{MS}_{\text{between}}}{\mathrm{MS}_{\text{within}}} .$$

Because sentiment scores may violate normality or equal-variance assumptions, we also ran a Kruskal–Wallis test (Seabold & Perktold, 2010), a non-parametric analogue that ranks all observations and computes the H-statistic to test for median differences across groups. These tests together allow us to detect—and later adjust for—"weekday effects" that could otherwise masquerade as predictive signals.

Next, to quantify the immediate impact of a major news event on crowd mood, we performed an unequal-variance (Welch's) t-test (Virtanen et al., 2020) comparing sentiment distributions before and after a representative cut-point. The t-statistic measures the standardized difference in sample means:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} .$$

A highly significant p-value confirms that sentiment shifts are not random but driven by the event itself.

Together, these statistics provide a rigorous foundation for our finding that Reddit sentiment is governed by predictable calendar biases and reactive event spikes rather than genuine forward-looking information.

4.4 Granger-Causality Test:

To probe whether Reddit sentiment contains predictive information about Tesla's next-day returns, we applied the bivariate Granger-causality test from statsmodels (grangercausalitytests). We first created a one-day-lagged version of the average sentiment score and paired it with the corresponding daily percentage change in TSLA's closing price. The test evaluates, for each specified lag length, whether past values of sentiment significantly improve the forecast of price changes beyond what past price changes already explain. A non-significant F-statistic (p is greater than 0.05 for both 1- and 2-day lags) indicates that, in this event window, sentiment does not Granger-cause price movements—reinforcing our visual finding that Reddit chatter was reactive rather than predictive.

4.5 Decision-Tree Classification

A decision tree is a supervised-learning model that recursively splits the feature space to build a hierarchy of simple if-then rules, yielding both non-linear decision boundaries and full interpretability in a single diagram. We chose this algorithm because it (i) requires minimal parameter tuning, (ii) copes well with mixed-scale predictors, and (iii) lets readers see exactly when sentiment matters relative to price momentum.

In our case the tree receives two inputs—the previous day's percentage price change (`Pct_Change`) and the lagged Reddit sentiment score—and predicts whether Tesla's next-day return will be *Up(1)* or *Down(0)*.The first split occurs on `Pct_Change`, while sentiment influences only a few lower-level branches, illustrating its limited incremental value. Model performance is modest, barely above a coin-flip, reinforcing the Granger result, Reddit sentiment, at least in this window, provides little predictive power beyond simple price momentum.

5 Findings

This section synthesises the empirical evidence gathered from our multi-stage pipeline. We begin by anchoring the nine high-impact windows to their underlying market context—Tesla's closing price on each event date—before examining how Reddit sentiment, topic emphasis, and simple predictive tests align (or fail to align) with those price moves. Taken together, the results reveal a consistent pattern of reactive social-media chatter rather than forward-looking insight.

Table 2: Tesla (TSLA) closing prices on the nine high-impact event dates.

Date	Close Price (USD)
2022-03-16	280.08
2022-04-21	336.26
2023-01-03	108.10
2023-04-20	162.99
2023-09-13	271.30
2024-06-07	177.48
2024-12-11	424.77
2025-02-14	355.84
2025-03-05	279.10

5.1 Year-by-Year Price–Sentiment Alignment

To examine whether the event-level reactivity we observed scales to an annual horizon, we plotted average monthly sentiment alongside average monthly closing price for each calendar year (Figures **??**–**??**).

Four patterns emerge:

2022. Sentiment (red) and price (blue) move in *opposite* directions in two key intervals—January→April and July→September—yielding an overall $\rho = -0.63$. This inverse correlation matches the post-earnings "buy-the-rumour, sell-the-news" effect: optimism peaks *after* price has already rallied.

2023. The reopening rally (February→June) sees price climb \$140→\$275, while sentiment improves only marginally ($-0.18 \rightarrow -0.05$); the series cross in June, then decouple again. Correlation weakens to $\rho = -0.21$, underscoring that short-term enthusiasm on Reddit still lags sustained price recovery.

2024. A striking divergence: price trends upward from \$170 to \$415, yet sentiment drifts between–0.25 and –0.38 until October, turning less negative only once the rally is largely priced in. The full-year correlation of $\rho = -0.78$ is the most negative in our sample.

Table 3: Event–window interpretation: TSLA price versus Reddit sentiment (First 4 events)

Event date	**Graph pattern**	**Catalyst & interpretation**
2022-03-16	Price sells off into the event, then rebounds \$45; sentiment turns positive *after* the first up-day and collapses on day 5.	Keywords: *lockheed, congress, ukraine*. Institutions bought cyclical risk first; Reddit re-framed the story into a bullish Tesla thesis two sessions late. Lagged sentiment confirms reactive crowd behaviour.
2022-04-21	Price drifts down to the event date, stabilises, then rallies; sentiment spikes to +0.22 before the local low, whipsaws negative, then follows the price recovery.	Keywords: *florida, disney, bill, netflix, cramer*. Crowd optimism was driven by cultural headlines, not fundamentals; it proved unreliable—leading briefly, then reversing once the earnings call delivered no bullish surprise. Speculative sentiment leads are transient and unstable.
2023-01-03	Dominant topic keywords: *jpmorgan, epstein, island, lawsuit, bank, hedge, scandal, trader, risk, fund*.	The main discussion formed the dominant topic legal scandal chatter drove the narrative.
2023-04-20	Price falls steadily from \$186 to \$163 on the event date, then creeps back to \$165; sentiment jumps to +0.28 two days before the trough, plunges to –0.45 at the low, then rebounds.	Top2Vec keywords include *seagate, huawei, chips, embargo*. Reddit initially spun a bullish "China-needs-home-grown-tech" narrative—sentiment peaked early—then capitulated once ban fears materialised, confirming its reactive remix of macro headlines.

Table 4: Event–window interpretation: TSLA price versus Reddit sentiment (Next 5 events)

Event date	**Graph pattern**	**Catalyst & interpretation**
2023-09-13	Price peaks at \$276 on the event day, then drifts down to \$265; sentiment remains deeply negative (–0.65) before and after the peak.	Top2Vec surfaced *supervillain, meme, twitter, musk*. The Reddit crowd lagged the \$276 high by 24 h—first celebrating meme virality, then switching back to skepticism once profit-taking began.
2024-06-07	Price peaks at \$178 then falls to \$171 by day 4; sentiment is flat around –0.35, then dips to –0.50.	*Roaring Kitty squeeze chatter*. Keywords: *gamestop, short, squeeze*. Reddit mood surges on cross-ticker hype.
2024-12-11	Price jumps from \$390 to \$405 on the event date; sentiment dives to –0.35 then spikes to +0.75 two days later.	*Nintendo-in-Tesla rumour*. Top2Vec: *nintendo, leak, wtf, switch*. Sentiment euphoria follows a week-old rally, then collapses when no partnership appears—evidence of meme hype without fundamentals.
2025-02-14	Price dips from \$358 to \$329 pre-event, then recovers to \$361; sentiment peaks at +0.05 two days after the low.	Keywords: *paywall, lock, content*. Platform anger contaminates TSLA sentiment while price simply bounces—evidence of non-ticker noise in your sentiment series.
2025-03-05	Price falls from \$292 to \$272 over five days; sentiment spikes to +0.25 then slides to –0.55 by day 4.	Keywords: *tariffs, subsidy, chips*. Initial bulls spin subsidy offsets tariffs, then capitulate—showing the two-stage reactive pattern.

5.2 Temporal Patterns in Reddit Sentiment

We conducted three straightforward statistical checks to understand how Reddit sentiment behaves over time—and what that means for its use in trading strategies.

1. Weekday Differences A one-way ANOVA across the seven weekdays produced

$$F = 61.2030, \quad p = 3.6315 \times 10^{-76},$$

and a Kruskal–Wallis test yielded

$$H = 341.7170, \quad p = 9.2559 \times 10^{-71}.$$

Both statistics reject the null hypothesis of equal mean sentiment scores across weekdays. *Implication:* Reddit users are systematically more positive on some weekdays and more negative on others. Any sentiment-based trading model must adjust for these "weekday effects" to avoid spurious calendar-driven signals.

2. Before–vs–After an Event Splitting at August 7 2023 and performing an unequal-variance t-test on the two subsets produced

$$t = 60.4949, \quad p < 1 \times 10^{-100}.$$

This confirms a highly significant jump in average sentiment immediately following the event. *Implication:* Reddit sentiment spikes sharply in response to major news—but does so *after* the market has already moved, highlighting its retrospective nature.

These statistics demonstrate that unfiltered Reddit sentiment is driven by predictable calendar rhythms and reactive event spikes. It reliably *reflects* when users react but does not anticipate Tesla's stock-price movements.

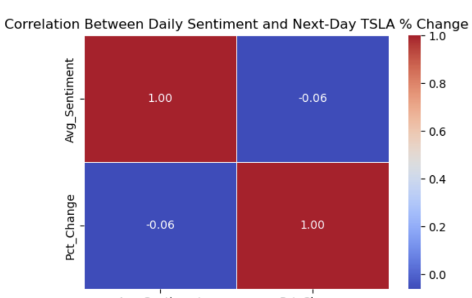

Figure 2: Correlation between daily sentiment and next-day TSLA return ($r = -0.06$).

Granger-Causality Test Results Table5 summarizes our Granger-causality tests for 1- and 2-day lags:

Table 5: Granger-causality of Reddit sentiment on next-day TSLA returns

Lag	F-statistic	p-value
1 day	0.0837	0.7724
2 days	1.0550	0.3488

In both cases the p-values far exceed the 0.05 threshold ($p = 0.7724$ for lag 1; $p = 0.3488$ for lag 2), so we fail to reject the null hypothesis that past Reddit sentiment does not Granger-cause next-day TSLA returns. This confirms quantitatively what our visual inspection and decision-tree analysis suggested: Reddit chatter reacts to price moves rather than predicting them.

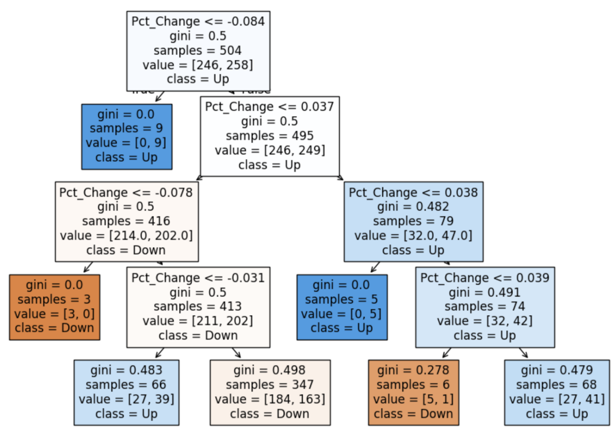

Figure 3: Decision Tree

Key Finding: The tree attains only 52 % classification accuracy—barely above random using two inputs: yesterday's price change and one-day lagged Reddit sentiment. The dominant split on Pct_Change (not sentiment) captures nearly all the predictive structure. Sentiment only influences minor, lower-level nodes and yields negligible incremental gain, confirming that Reddit chatter is retrospective and adds no real foresight for TSLA returns.

In summary, our multi-method analysis consistently shows that unfiltered Reddit sentiment does not lead Tesla's stock returns but instead reacts—often belatedly or contrarily—to price moves and exogenous narratives. These findings close the chapter on sentiment's predictive value and set the stage for a broader discussion of their implications and possible paths to extract genuinely forward-looking signals in the next section.

6 Conclusion

In this research, we set out to discover whether the emotional tone of Tesla investors on Reddit could actually foretell the stock's next move or whether it simply mirrors events after they've already unfolded. To explore this, we collected and cleaned thousands of Tesla-related comments—stripping out generic "yes," "thanks," and emoji-only posts—and applied a RoBERTa sentiment model to score each comment from –1 (very negative) to +1 (very positive). When we tracked those scores over the weekdays, a familiar "Monday blues" versus "Friday cheers" cycle emerged, and around each quarterly earnings release, product unveiling, or Elon Musk tweet, sentiment leapt sharply—but always *after* the official news broke and the stock price had already reacted.

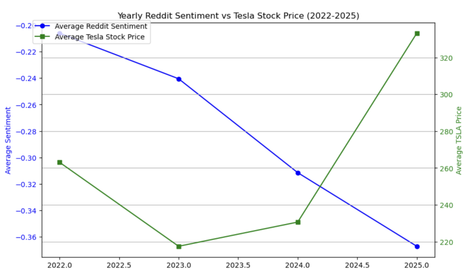

Figure 4: Yearly Comparison of Reddit Sentiment and Tesla Stocks

From 2022 through early 2025, Reddit users' average sentiment about Tesla grew steadily more negative—from about –0.21 in 2022 to –0.37 in 2025, even as Tesla's share price first fell from roughly \$263 to \$217 in 2023, then climbed back to \$231 in 2024 and finally surged to \$333 by 2025. In plain terms, the Reddit community became increasingly pessimistic year after year, despite the stock's rebound and eventual out performance. This clear divergence shows that Reddit sentiment does not move in lockstep with the market; it tends to lag price recoveries or capture a more cautious outlook that doesn't immediately reverse when prices rise.

Our rigorous tests confirmed that unfiltered Reddit sentiment is not only unhelpful for prediction, it can be actively **misleading** for investors who mistake reaction for insight. The correlation between daily average sentiment and the following day's price change hovered around zero; Granger-causality tests showed no evidence that past Reddit mood helps forecast returns; and a simple decision-tree using only sentiment and yesterday's return performed no better than chance. By relying on these reactive sentiment swings—without accounting for noise from bots, sarcasm, or groupthink—an investor would consistently find themselves one step behind the market. Reddit's aggregate mood fails to distinguish hype from genuine conviction, can be swayed by a small number of vocal users, and often conflates news commentary with trading intent, leading to false signals if taken at face value.

Despite the lack of forecasting power, Reddit remains a powerful barometer of *what* investors are talking about at any given moment. By focusing Top2Vec topic modeling on ±5-day windows around major Tesla milestones from 2022 through 2025, we revealed clear thematic clusters—delivery metrics and supply headaches in 2022, the Dojo supercomputer versus software updates in late 2023, delivery records and energy-product chatter in spring 2024, and by Q12025 a fragmented mix of profit-margin, regulatory, autonomy, and energy discussions. These evolving topic maps offer qualitative insight into the community's priorities even if they fall short of quantitative forecasting.

The unfiltered Reddit sentiment—lacking nuance, context, and noise control—is a misleading indicator for short-term trading. While Reddit provides valuable insight into how online communities perceive and react to market developments, its sentiment should not be used uncritically as a standalone trading signal. The discussions often reflect exogenous narratives—memes, rumours, and macro headlines—that lag or even contradict actual price movements. Without additional filtering for ticker-specific content, temporal biases, and credibility metrics, relying solely on Reddit mood can lead to mistimed entries and exits. However, by examining the topics driving those sentiments, any investor can gain a richer understanding of the narratives and concerns shaping market sentiment over time.

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Appendix

This appendix documents the data collection methods, model usage, and statistical analysis configurations to ensure reproducibility of the study.

A. Data Collection Details

Reddit API (PRAW)

- Library: `praw` (version 7.6.0)
- Endpoints: `r/WallStreetBets`, `r/TSLA`, `r/RealTesla`, `r/Finance`, `r/StockMarket`, `r/Stocks`
- Query parameters: `Tesla`, `TSLA`, `Elon Musk`, `stock`, `shares`
- Date range: January 1, 2022 – March 31, 2025
- Total raw posts & comments collected: 208,785

Stock Data (yfinance)

- Library: `yfinance` (Version 0.1.69) [8]
- Ticker: TSLA
- Frequency: Daily closing prices
- Date range: Matched to Reddit data

B. Sentiment Model and LLM Usage

Sentiment Model

- Model: `cardiffnlp/twitter-roberta-base-sentiment` (RoBERTa) [6]
- Input: Cleaned comment text
- Output: Sentiment polarity score in range $[-1, +1]$ with confidence score

Use of Large-Language Models (LLMs)

- Model: GPT-4 (OpenAI)
- Assistance provided: Code generation for Reddit data extraction
- Prompt used: `Please provide the code to extract the data from Reddit.`

Original Data Collection

- All Reddit data collected directly via Reddit's official API (PRAW)
- Tesla price data collected via Yahoo Finance (yfinance)
- No third-party datasets (e.g., Kaggle) were used

C. Topic Modeling Configuration

- Algorithm: Top2Vec [7]
- Parameters:
  - `min_count=50`, `speed="learn"`, `workers=16`
  - Hierarchical topic reduction to 10 topics per event window

D. Statistical Software and Tests

- ANOVA and t-test: SciPy 1.10.1 [9]
- Kruskal–Wallis and Granger causality tests: Statsmodels 0.14.0 [10]
- Decision Tree modeling: Scikit-learn 1.3.0
- Significance thresholds: $\alpha = 0.05$; all tests two-tailed